1777–78

1779–80

❝ MY **SIGHT BECOMES DISORDERED…** LIKE A PERSON WHO HAS LOOKED AT THE SUN… THE **PAIN IN MY HEAD** COMMENCES… WITH **GREAT SEVERITY…** ❞

Samuel-Auguste Tissot, Swiss physician, record of a patient's symptoms from *Treatise on the Nerves and Nervous Disorders*, 1783

Designed by Thomas Pritchard and built by Abraham Derby III, the bridge at Ironbridge, which crosses the Severn River in Shropshire, England, was fabricated entirely from precast pieces of iron.

MATHEMATICIANS HAD LONG BEEN BAFFLED BY THE PROBLEM OF finding the roots of negative numbers, calling them "imaginary numbers." Then, in 1777, Swiss mathematician **Leonhard Euler** introduced the imaginary unit, the **symbol *i***, which gives –1 when squared. Euler's insight meant the square root of any negative number could be included in equations as *i* times the square root of the number.

In 1777, London clockmaker **John Arnold** (1736–99) created a **watch of unprecedented accuracy** for calculating longitude at sea, improving on Harrison's H4 of 1759. Arnold named such timepieces "chronometers."

Swiss physician **Samuel-Auguste Tissot** (1728–97) described **migraine**. Although Tissot wrongly thought migraines began in the stomach, he described the symptoms very accurately—the severity of the pain, its recurrence, the suddenness of onset, the effect on vision, and vomiting.

Scottish surgeon **John Hunter** (1728–93) wrote an important **study of teeth**. He also advocated transplanting good teeth from donors to replace rotten ones, but transplanted teeth were rejected by the recipient's immune system.

Tooth transplant
This cartoon satirizes John Hunter's practice of buying healthy teeth from the poor to implant in place of lost teeth in the mouths of the rich.

WITH A SIMPLE EXPERIMENT IN 1779, Dutchborn physician **Jan Ingenhousz** (1730–99) discovered the essence of **photosynthesis**, the chemical reaction by which plants make food from sunlight (see 1783–88). When Ingenhousz set plants under water in a glass container, he saw gas bubbles form on the undersides of leaves, which he showed was oxygen. But the bubbles formed only in sunlight, not in darkness. Ingenhousz found that **plants need sunlight for respiration**—when they take in gases from the air to make glucose for energy—and oxygen is released as a waste product.

Factory production of cotton was accelerated in 1779 when English inventor **Samuel Crompton** (1753–1827) created an **ingenious hybrid, or mule**, combining a spinning machine with a weaving machine to produce finished cloth from raw thread. Also in 1779, large-scale construction using iron began with the **bridge at Ironbridge** in England, designed by **Thomas Pritchard** (*c.*1723–77) and built by Abraham Derby III (1750–91).

In 1780, in the memoir *On Combustion in General*, French chemist **Antoine Lavoisier** finally demolished the theory that burning materials lose a substance called phlogiston. By meticulous weighing before and after burning, Lavoisier showed that substances can gain weight when they burn—by combining with oxygen in the air—not lose it. Together with **Pierre-Simon Laplace** (1749–1827), a French mathematician, Lavoisier invented the **ice calorimeter** for measuring the heat produced by chemical changes. This was the beginning of the science of thermochemistry. With his ice calorimeter, Lavoisier demonstrated that animals produce heat without losing weight, so disproving the

Lavoisier–Laplace calorimeter
Heat produced by a mouse placed in the inner chamber was measured by the volume of water produced by melted ice in the outer chambers.

1777 Leonhard Euler introduces the imaginary unit *i*.

1777 John Arnold introduces the term "chronometer"

1777 German–Swedish chemist **Carl Scheele** discovers **molybdenum**

1778 English geographer **James Rennell** charts the Agulhas Current

1778 German anatomist **Samuel von Sömmerring** identifies the 12 **cranial nerves**

1778 Samuel-Auguste Tissot describes **migraine**

1778 John Hunter publishes The Natural History of the Human Teeth

1779 Jan Ingenhousz discovers plant **photosynthesis**

1779 Construction of the **first all-iron bridge** begins at Ironbridge, Shropshire, in England

1779 Samuel Crompton invents the spinning **mule**

1780 Antoine Lavoisier demonstrates the role of oxygen in combustion

1780 Lavoisier and Pierre-Simon Laplace invent the **ice calorimeter**